Document Structure

LaTeX: A Professional Document Preparation System

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November 2026



Agenda

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    - 1.2 Numbered Sections
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Introduction

Documents usually have some form of “logical structure”: division into

- chapters,
- sections,
- sub-sections,
- etc.,

to organize their content.

LaTeX supports the creation of a document structure and also enables customization of sectioning and numbering. The commands available to organize a document depend on the document class being used, although the simplest form of organization, sectioning, is available in all formats.

Sectioning

```
\documentclass{article}
\usepackage{blindtext}

\title{Sectioning}
\author{John Doe}
\date{\today}

\begin{document}
\maketitle
\section{Introduction}
This is the first section. \blindtext

\section{Second Section}
This is the second section. \blindtext
\end{document}
```

Output

Sectioning

John Doe

December 25, 2022

1 Introduction

This is the first section. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

2 Second Section

This is the second section. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis

Level of Sectioning

LaTeX can organize, number, and index chapters and sections of document. There are 7 levels of depth for defining sections depending on the document class:

Level	Command
-1	`\part{PartName}`
0	`\chapter{ChapterName}`
1	`\section{SectionName}`
2	`\subsection{SubSectionName}`
3	`\subsubsection{SubSubSectionName}`
4	`\paragraph{ParagraphName}`
5	`\subparagraph{SubParagraphName}`

Numbered Sections

```
1  \documentclass{article}
2  \usepackage{blindtext}
3
4  \title{Sectioning}
5  \author{John Doe}
6  \date{\today}
7
8  \begin{document}
9  \maketitle
10 \section{Introduction}
11 This is the first section. \blindtext
12 \subsection{A Minor Topic}
13 This is a subsection. \blindtext
14
15 \subsection{Another Minor Topic}
16 This is also a subsection. \blindtext
17
18 \section{Second Section}
19 This is the second section. \blindtext
20
21 \end{document}
```

Output

Sectioning

John Doe

December 25, 2022

1 Introduction

This is the first section. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

1.1 A Minor Topic

This is a subsection. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

1.2 Another Minor Topic

This is also a subsection. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo,

Table of Contents

```latex
\documentclass{article}
\usepackage{blindtext}

\title{Sectioning}
\author{John Doe}
\date{\today}

\begin{document}
\maketitle
\tableofcontents
\section{Introduction}
This is the first section. \blindtext
\subsection{A Minor Topic}
This is a subsection. \blindtext
\subsection{Another Minor Topic}
This is also a subsection. \blindtext
\section{Second Section}
This is the second section. \blindtext
\subsection{Second Minor Topic}
This is a subsection. \blindtext
\subsection{Futher Minor Topic}
This is also a subsection. \blindtext
\end{document}
```

Output

Sectioning

John Doe

December 25, 2022

Contents

1 Introduction **1**
1.1 A Minor Topic 1
1.2 Another Minor Topic 2

2 Second Section **2**
2.1 Second Minor Topic 2
2.2 Futher Minor Topic 3

1 Introduction

This is the first section. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent

Unnumbered Sections

```
1  \tableofcontents
2  \section*{Introduction}
3  This is the first section. \blindtext
4
5  \subsection{A Minor Topic}
6  This is a subsection. \blindtext
7
8  \subsection{Another Minor Topic}
9  This is also a subsection. \blindtext
10
11 \section{Second Section}
12 This is the second section. \blindtext
13
14 \subsection*{Second Minor Topic}
15 This is a subsection. \blindtext
16
17 \subsection{Futher Minor Topic}
18 This is also a subsection. \blindtext
```

Output

Sectioning

John Doe

December 25, 2022

Contents

0.1 A Minor Topic 1
0.2 Another Minor Topic 2

1 Second Section **2**
1.1 Futher Minor Topic 3

Introduction

This is the first section. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

0.1 A Minor Topic

The article class

```
1  \documentclass{article}
2  \usepackage{blindtext}
3  \title{Sections in an Article}
4  \author{John Doe}
5  \date{\today}
6  \begin{document}
7  \maketitle
8  \tableofcontents
9
10 \section{What is it and what makes it so different?}
11 Here is a first section. \blindtext
12
13 \subsection{Explaining: Where to start?}
14 This is a subsection. \blindtext
15
16 \section{Background}
17 This is a section. \blindtext
18
19 \subsection{Opening up the black box}
20 This is a subsection. \blindtext
21 \end{document}
```

Output: Article

Sections in an Article

John Doe

25 December 2022

Contents

1 What is it and what makes it so different? 1
1.1 Explaining: Where to start? 1

2 Background 2
2.1 Opening up the black box 2

1 What is it and what makes it so different?

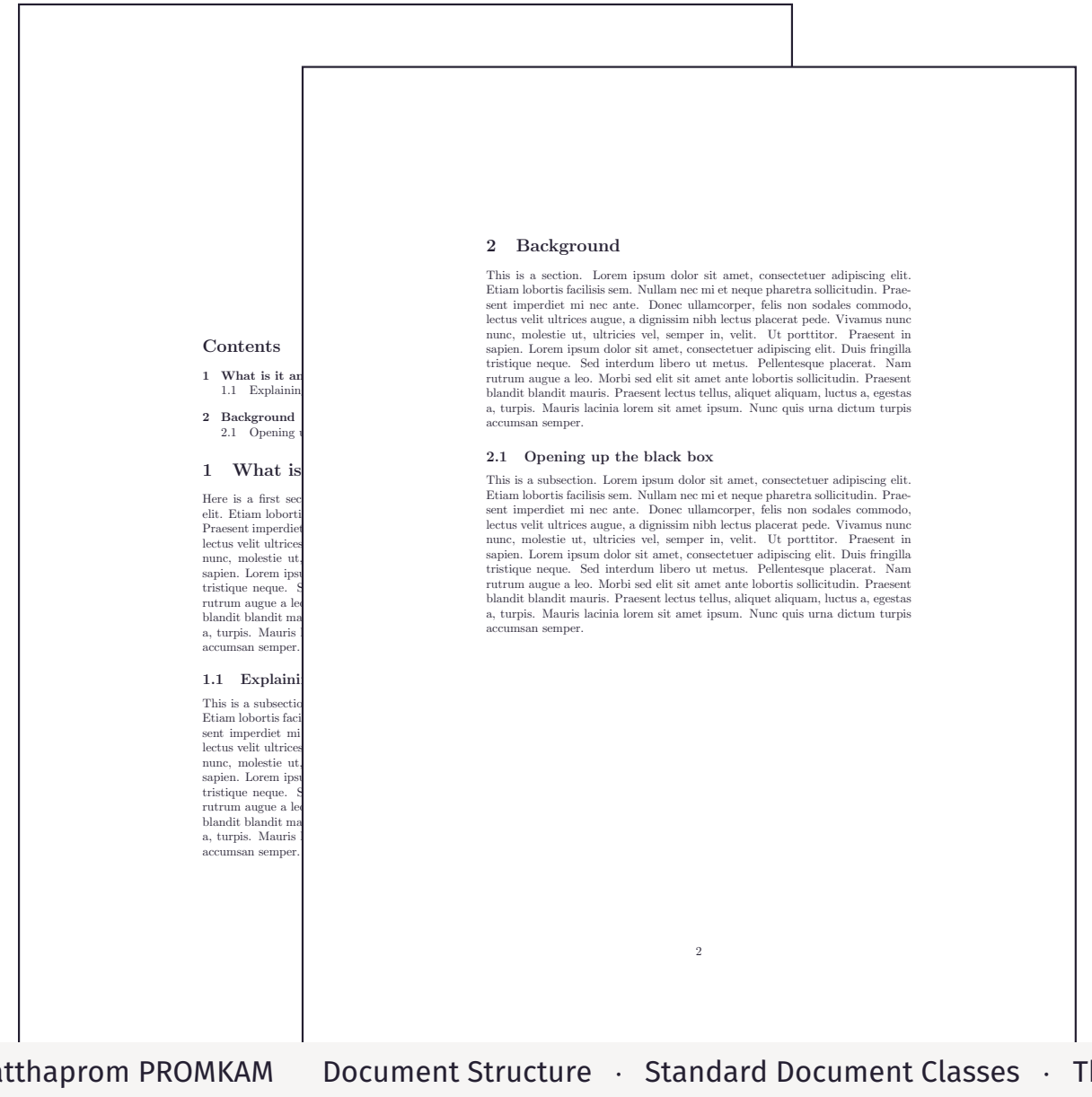
Here is a first section. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

1.1 Explaining: Where to start?

This is a subsection. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

1

Output: Article



The report class

```
\documentclass{report}
\usepackage{blindtext}
\title{Sections and Chapters}
\author{John Doe}
\date{\today}
\begin{document}
\maketitle
\tableofcontents

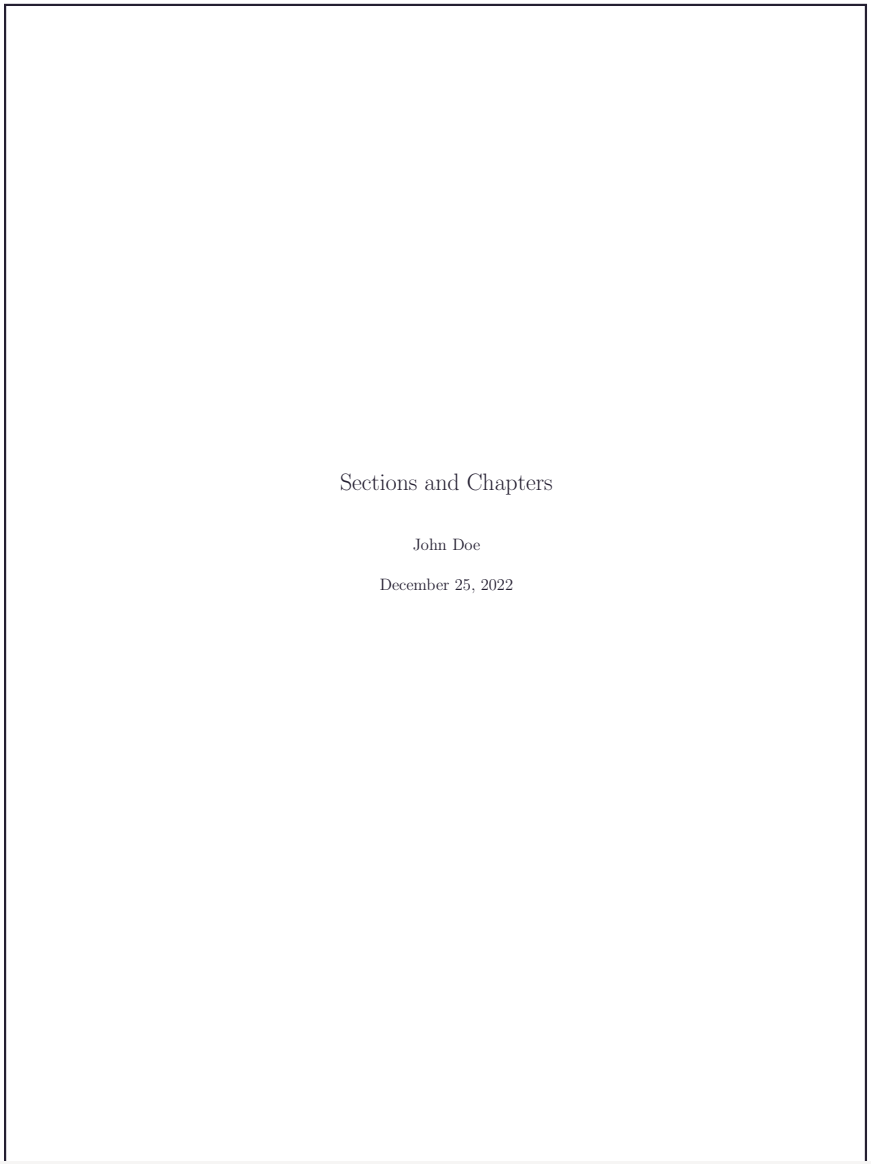
\chapter{An Introduction }
\section{What is it and what makes it so different?}
Here is a first section. \blindtext

\subsection{Explaining: Where to start?}
This is a subsection. \blindtext

\chapter{Background and history}
\section{Background}
This is a section. \blindtext

\subsection{Opening up the black box}
This is a subsection. \blindtext
\end{document}
```

Output: Report



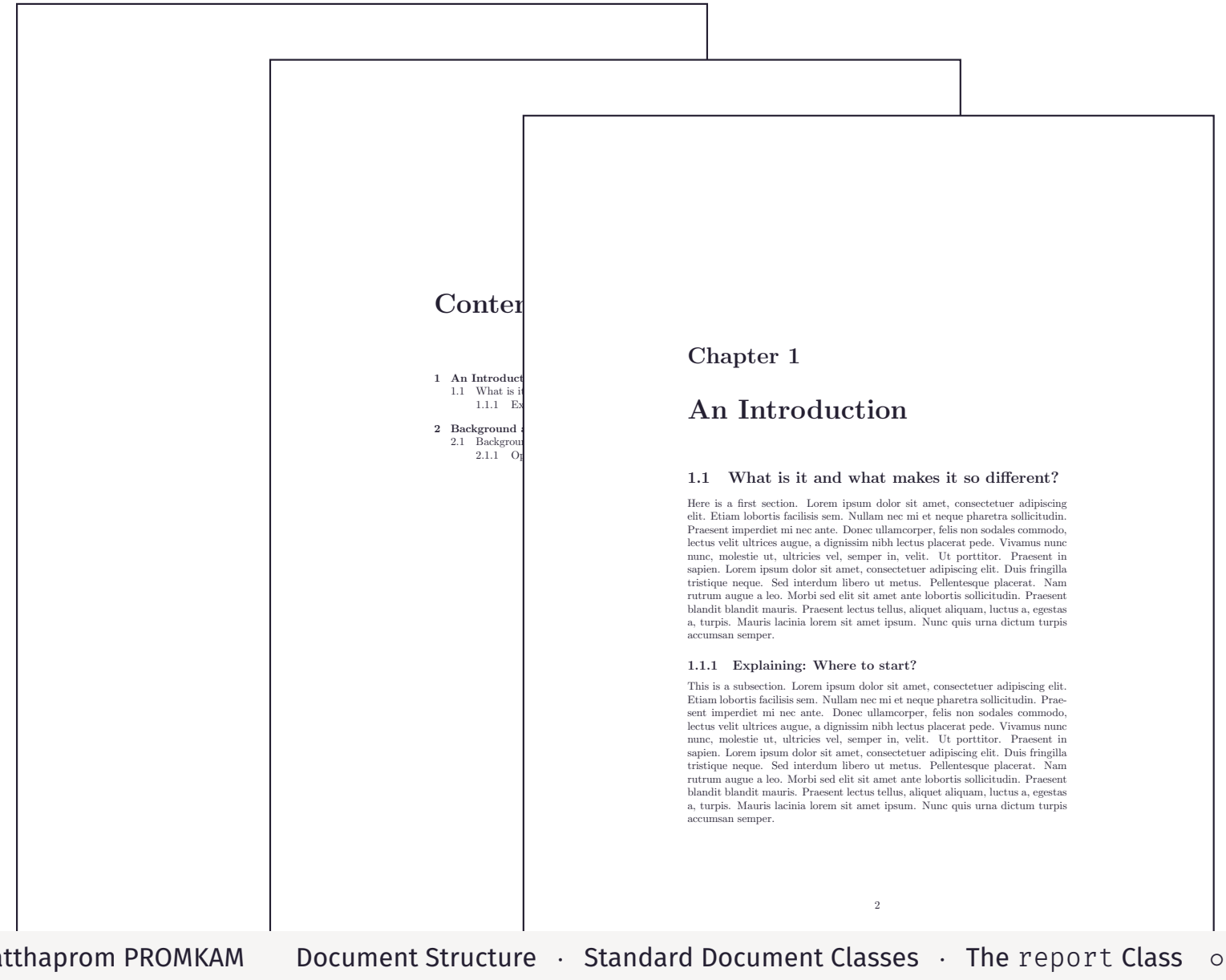
Output: Report

Contents

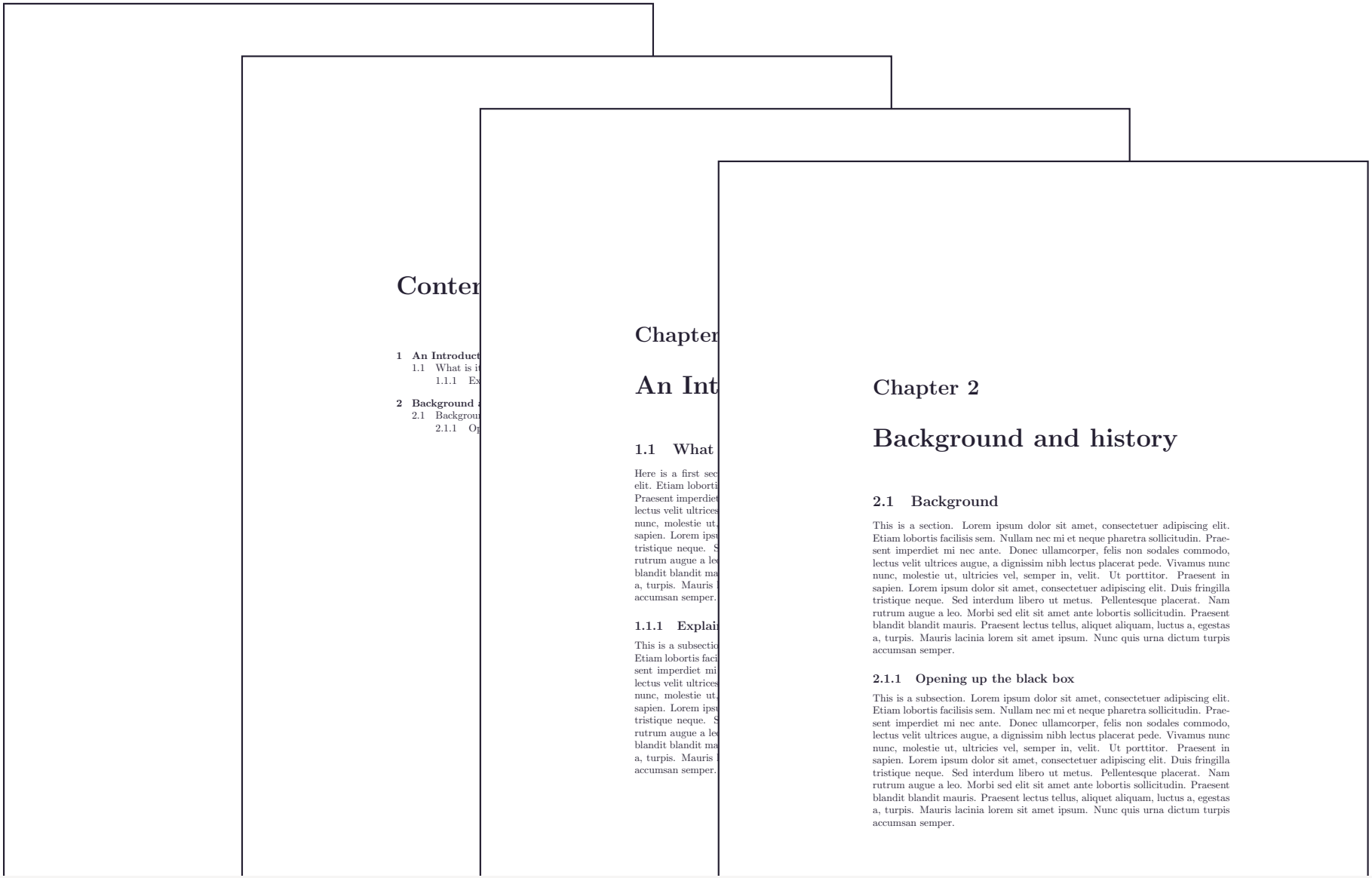
1 An Introduction **2**
1.1 What is it and what makes it so different? 2
1.1.1 Explaining: Where to start? 2

2 Background and history **3**
2.1 Background 3
2.1.1 Opening up the black box 3

Output: Report



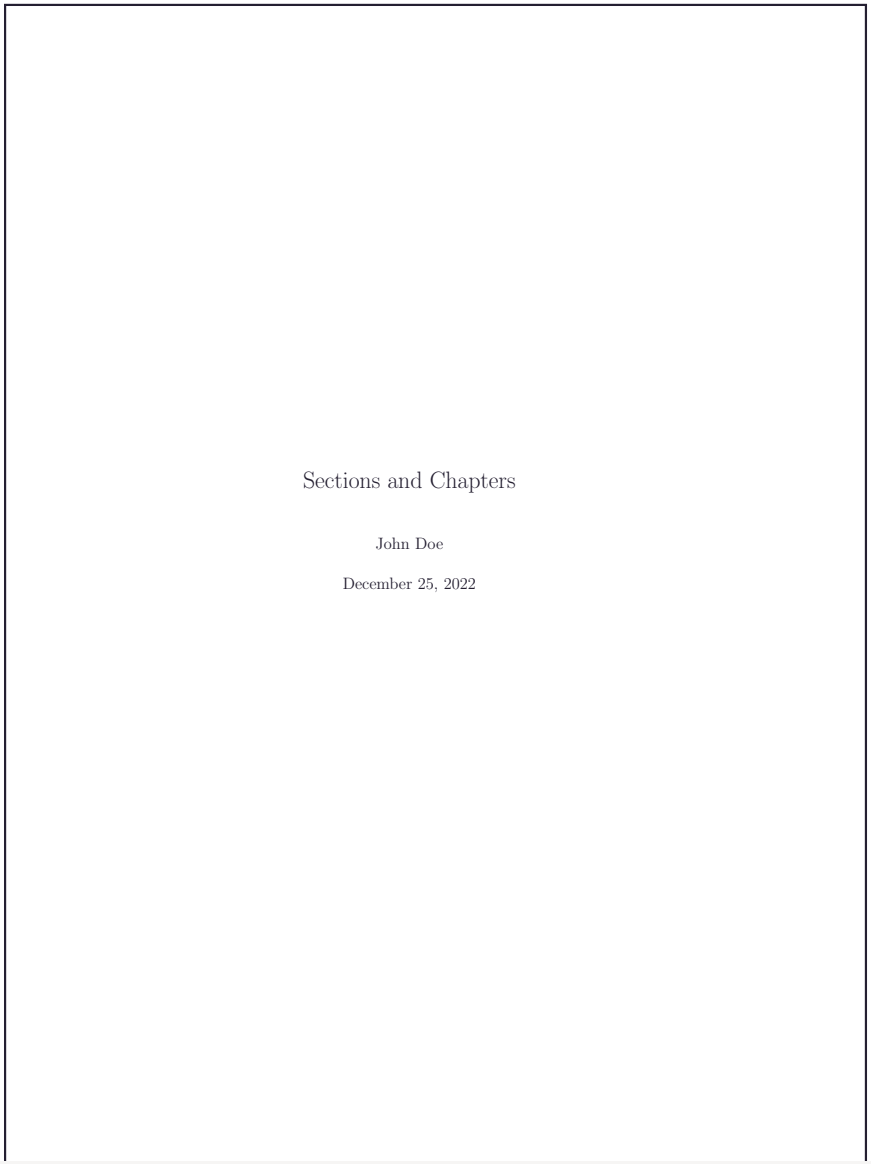
Output: Report



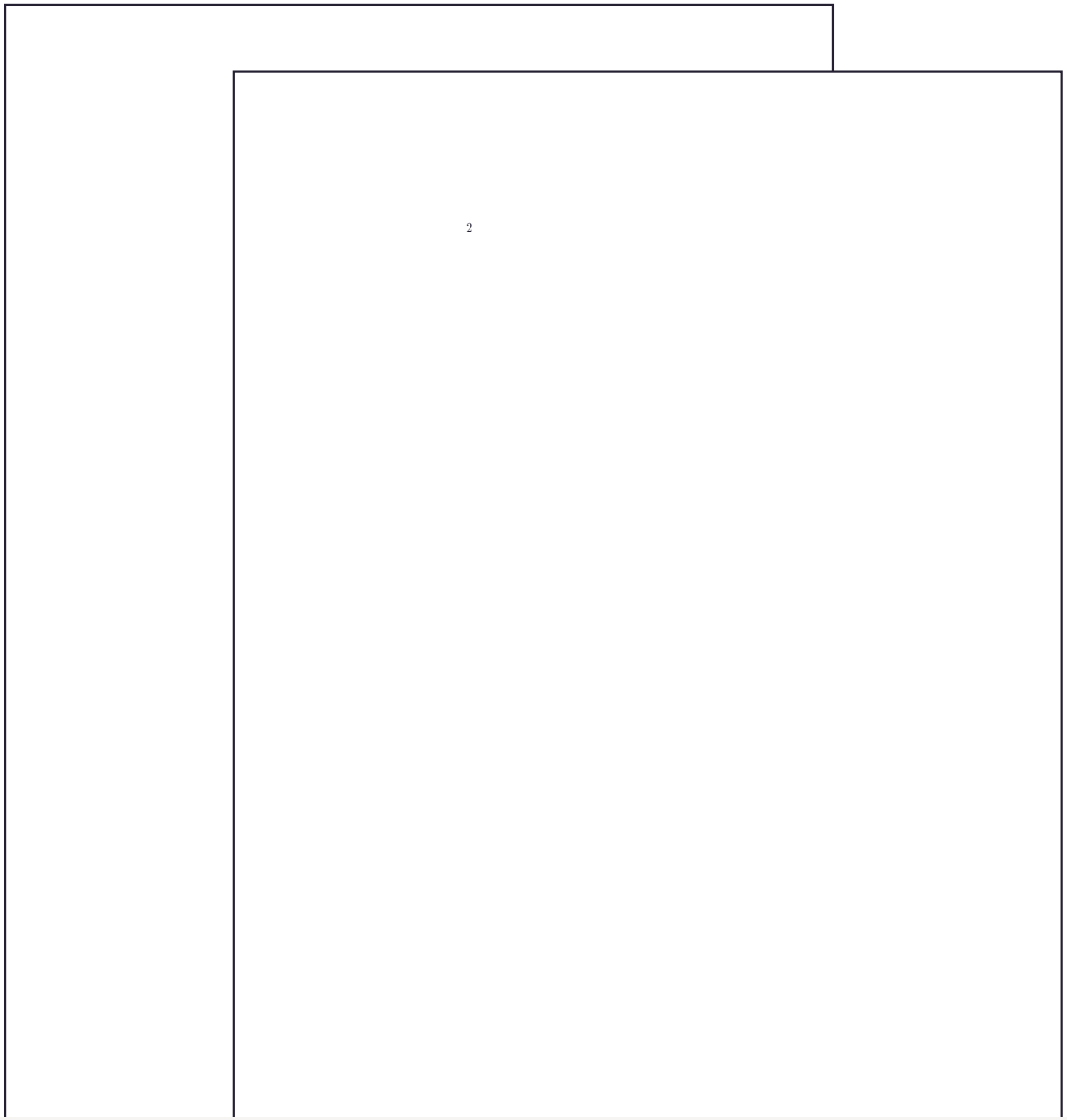
The book class

```
1  \documentclass{book}
2  \usepackage{blindtext}
3  \title{Sections and Chapters}
4  \author{John Doe}
5  \date{\today}
6  \begin{document}
7  \maketitle
8  \tableofcontents
9
10 \chapter{An Introduction }
11 \section{What is it and what makes it so different?}
12 Here is a first section. \blindtext
13
14 \subsection{Explaining: Where to start?}
15 This is a subsection. \blindtext
16
17 \chapter{Background and history}
18 \section{Background}
19 This is a section. \blindtext
20
21 \subsection{Opening up the black box}
22 This is a subsection. \blindtext
23 \end{document}
```

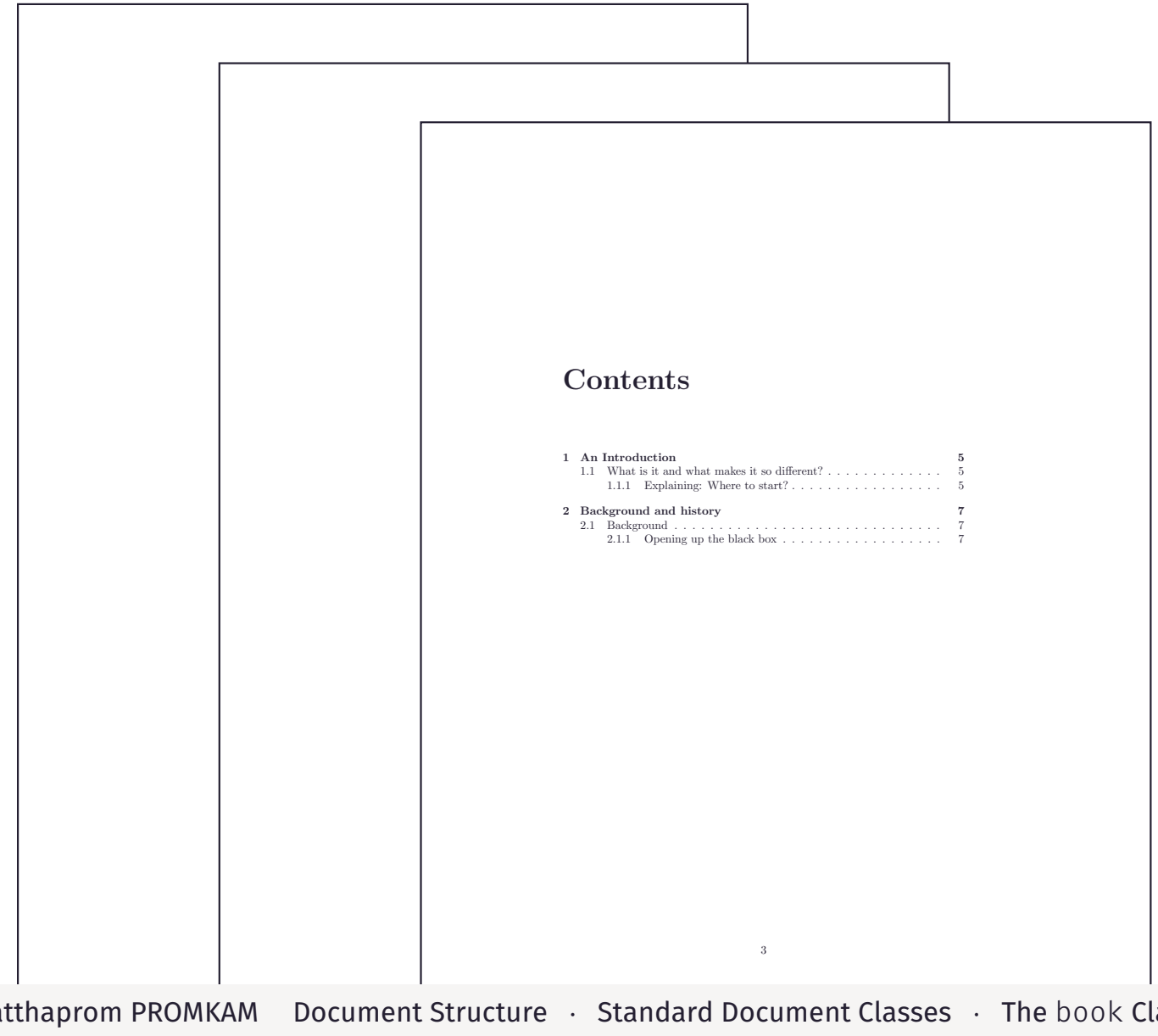
Output: Book



Output: Book



Output: Book



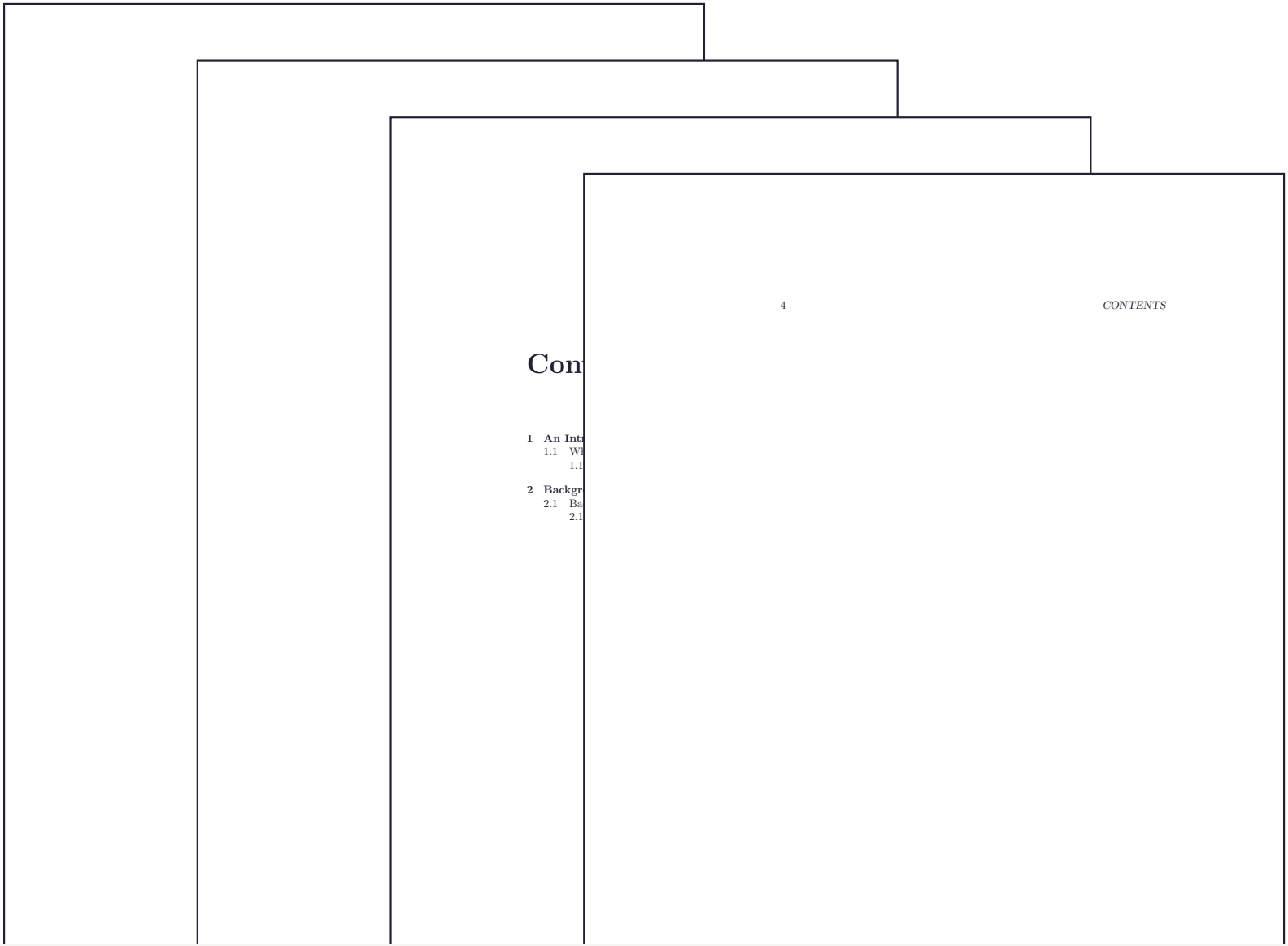
Contents

1 An Introduction **5**
1.1 What is it and what makes it so different? 5
1.1.1 Explaining: Where to start? 5

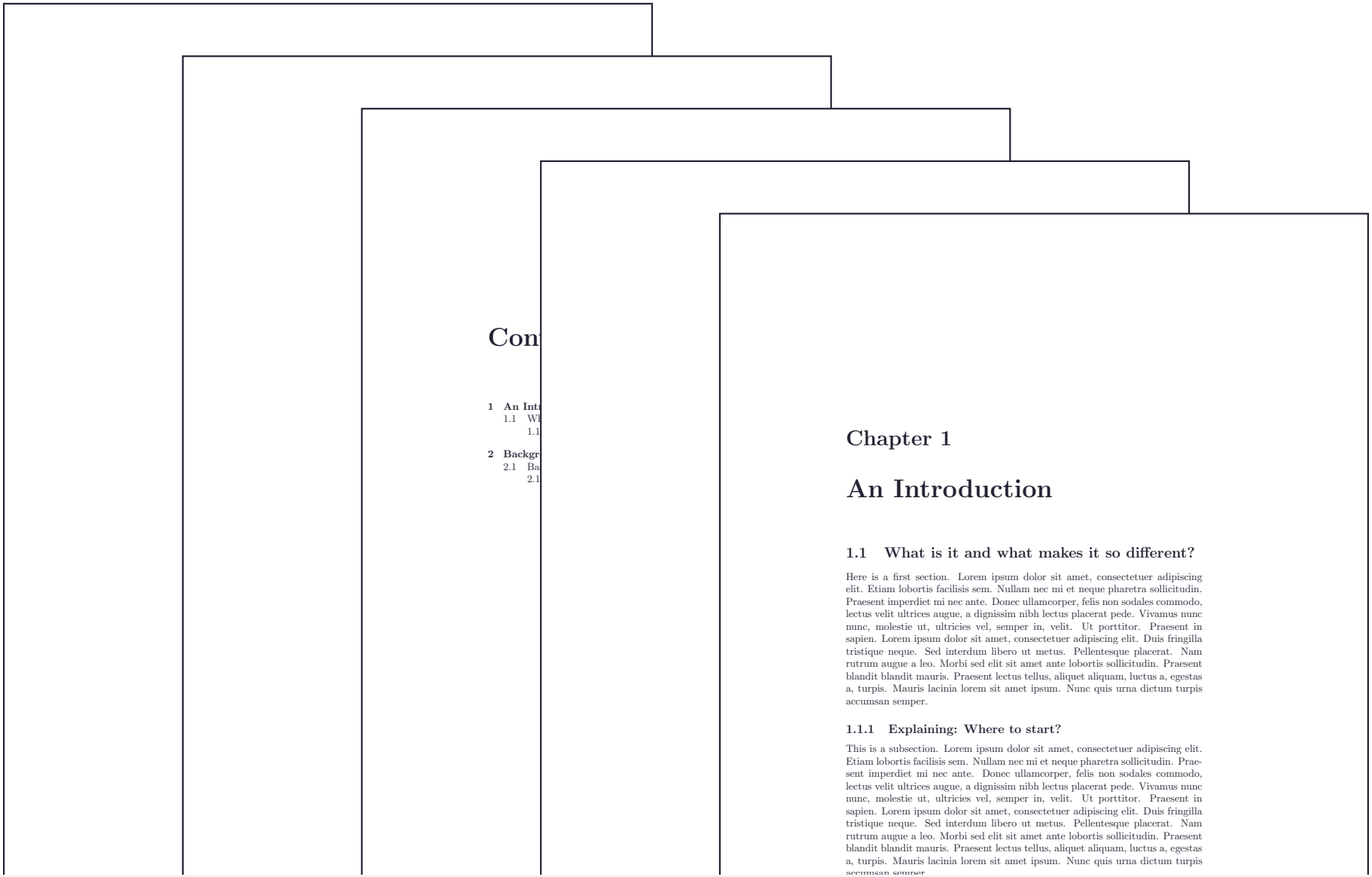
2 Background and history **7**
2.1 Background 7
2.1.1 Opening up the black box 7

3

Output: Book



Output: Book


Con
1 An Intr
1.1 Wh
1.1
2 Backgr
2.1 Ba
2.1
Chapter 1
An Introduction
1.1 What is it and what makes it so different?
Here is a first section. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.
1.1.1 Explaining: Where to start?
This is a subsection. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis

Document Classes: `article` vs. `report` vs. `book`

Differences with regard to **default settings**:

- The `book` class uses the twoside class option, while report and article use oneside.
- `book` uses openright, while `report` uses openany. (Note that "right" means an odd page in twoside mode, but any page in oneside mode.) For `article`, the distinction between openright and openany isn't applicable.
- `book` uses the headings pagestyle for non-chapter-starting pages, while `report` and `article` always use plain.
- `book` and `report` use titlepage, while `article` uses notitlepage.

Document Classes: `article` vs. `report` vs. `book`

Differences with regard to **default settings**:

- For `book` and `report`, the lowest-level sectioning command which is numbered and incorporated into the table of contents is `\subsection`, while for `article` it is `\subsubsection`.
- `book` and `report` will use the arguments of `\chapter` and `\section` for running headings, while `article` will use `\section` and `\subsections`.
- `book` and `report` will number floats (figures, tables etc.), equations, and footnotes per chapter, while `article` will number them continuously.
- `book` and `report` use `\bibname` for the heading of bibliographic references, while `article` use `\refname`.

Document Classes: `article` vs. `report` vs. `book`

Differences with regard to **available commands and environments**:

- `book` and `report` feature the `\chapter` sectioning command, while article doesn't.
- In `book` and `report`, `\appendix` will cause `\chapters` to be typeset as "Appendix X" instead of "Chapter X". For article, this isn't applicable.
- `book` and `report` will start a new page for `\parts` , while article won't.
- `book` offers the `\frontmatter`, `\mainmatter`, and `\backmatter` commands to control page numbering, while `report` and `article` don't.
- `book` doesn't offer the abstract environment, while `report` and `article` do.

Ratthaprom PROMKAM

RMUTT, November 2026

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